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Roll No

CS-7002-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

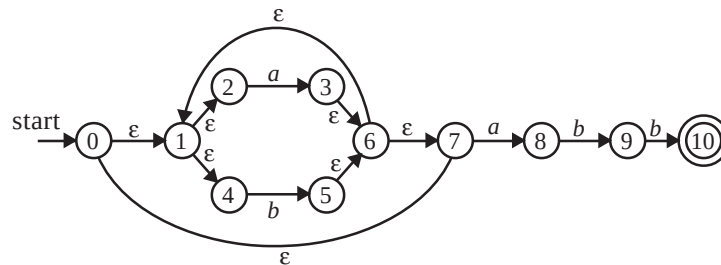
Compiler Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions (including all parts).
ii) All questions carry equal marks.
iii) Assume missing data, if any suitably.

1. a) Explain with neat diagram the various phases of compiler.
b) What is DFA? Construct the DFA machine for the given, with ϵ -NFA machine.



2. a) Describe the following in brief:
i) Ambiguity
ii) Left recursion
b) Consider the grammar:
 $E \rightarrow 5 + T \mid 3 - T$
 $T \rightarrow V \mid V * V \mid V + V$
 $V \rightarrow a \mid b$
What is the use of left factoring? Find the first and follow for above grammar and do the left factoring for it.

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3. a) Construct the SLR parsing table for the following grammar.
 $S \rightarrow L = R$
 $S \rightarrow R$
 $L \rightarrow R$
 $L \rightarrow id$
 $R \rightarrow L$
 b) Explain with an example, recursive descent parser with back tracking.
4. a) What are the features and capabilities of the symbol table?
 b) What are shift-reduce and reduce-reduce conflicts? Explain with suitable example.
5. a) Explain implicit type conversion and explicit type conversion with example.
 b) Explain different issues in the design of code generator.
6. a) Generate the three-address code for the following code segment:
 While ($a < c$ and $b < d$) do
 If $a = 1$ then $c = c + 1$;
 Else
 While ($a \leq d$) do
 $a = a + 3$;
 b) With an example, explain the various formats of intermediate code.
7. Explain various code optimization techniques in detail.
8. Write short notes on following: (Any Three)
 - a) Input Buffering
 - b) Canonical items
 - c) L-attributes and S-attributes
 - d) Back Patching

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